

ENGR 200 Fall 2019
Midterm 1 Study Questions

1. (a) If the occurrence of event B makes A more likely (i.e. $P(A|B) > P(A)$), then does the occurrence of event A make B more likely? Justify your answer.

(b) Suppose that E and F are events and $P(E) = 0.4$. What can you say about $P(E|F)$ if:
 - E and F are independent?
 - E and F are mutually exclusive?
 - $F \subset E$?
 - $E \subset F$?
2. A sample of 3 items is selected at random from a box containing 20 items of which 4 are defective. Let X define the random variable that specifies the number of defective items in the sample.

(a) Find the probability mass function (PMF) of X .

(b) What is the expected value of X ?

(c) What is the variance of X ?
3. Electronic components of a certain type are shipped to a supplier in batches of ten. There are three different types of batches: A, B and C. 50% of all batches are type A, 30% are type B, and 20% are type C. Batches of type A always have zero defective components (out of ten), batches of type B always have 1 defective component and batches of type C always have 2 defective components. Upon delivery, the quality control department selects a random batch and then randomly picks two components (without replacement) among the ten components in the batch.

(a) Given that the selected batch is type C, find the probability that exactly one of the two components selected by the quality control department is defective.

- (b) Find the probability that the exactly one of the two components selected by the quality control department is defective.
 - (c) Given that both of the components selected by the quality control department are defective, what is the probability that the selected batch was type C?
 - (d) Given that none of the two components selected by the quality control department is defective, what is the probability that the selected batch was type A?
 - (e) What is the expected number of defective components among the two components selected for inspection by the quality control department ?
4. You roll two dice repeatedly until getting a sum of 5 or 7 on the two dice. If you get a 5 before getting a 7, your payoff is 3 TL, otherwise your payoff is -2 TL. Answer parts a, b, c, d based on this game.
- (a) What is the expected number of tosses required until the end of the game (the game ends when you obtain a 5 or a 7)?
 - (b) What is the probability that your payoff at the end of the game is negative?
 - (c) What is the your expected payoff in the game?
 - (d) What is the variance of the payoff in the game?
5. You roll a single die 4 times. If the number that appears on the last roll is larger than all of the numbers that appear in the previous 3 tosses, your payoff is 100 TL, otherwise your payoff is -10 TL.
- (a) What is the probability that you win 100 YTL given that the number on the last roll is 3 ?
 - (b) What is the probability that you win 100 YTL in this game?
 - (c) What is the expected payoff in this game?

6. A quality control inspector is inspecting newly manufactured items for faults. The inspector searches an item for faults in a series of independent fixations (i.e. inspection stages). Given that a flaw is actually present, let p denote the probability that the flaw is detected during any one fixation.
- (a) Assuming that an item has a flaw, what is the probability that it is detected by the second fixation (once a flaw is detected, no more inspections are required)?
 - (b) Given an expression for the probability that a flaw will be detected by the end of the n 'th inspection.
 - (c) If the inspection line consists of three fixations, what is the probability that a flawed item will pass inspection?
 - (d) Suppose 10% of all items contain a flaw. With the assumption of part c, what is the probability that a randomly chosen item will pass inspection (note that it will automatically pass inspection if it is not flawed, but could also pass if it is flawed)?
 - (e) Given that an item has passed inspection, what is the probability that it is flawed (assume that $p = 0.2$)?
7. Assume A and B are independent events with $P(A) = 0.2$ and $P(B) = 0.3$. Let C be the event that at least one of A or B occurs, and let D be the event that exactly one of A or B occurs.
- (a) Find $P(C)$.
 - (b) Find $P(D)$.
 - (c) Find $P(A|D)$ and $P(D|A)$.
 - (d) Determine whether A and D are independent.
8. Given a class of 12 girls and 10 boys.
- (a) In how many ways can a committee of five consisting of 3 girls and 2 boys be chosen?
 - (b) What is the probability that a committee of five, chosen at random from the class, consists of three girls and two boys?

- (c) In how many ways can all the class sit around a table?
9. Let the probability mass function (pmf) of a random variable X , $p_X(x)$, be given as follows:
- $$p_X(x) = \begin{cases} a\sqrt{x} & x = 1, 4, 9 \\ 0 & otherwise \end{cases}$$
- (a) Find a to have a proper pmf.
- (b) Find $P(X = 4|X < 9)$.
- (c) Find the expected value of X .
- (d) Find the expected value of $\sqrt{X}$.
10. In several applications consumer demand is modeled as a Poisson random variable. Assume that you are trying to sell your old bicycle for 100 TL by hanging fliers around your dorm. You think that the demand for this bike will be a Poisson random variable with parameter $\lambda = 1$.
- (a) What is the probability that you are able to sell your bike? Note that to be able to sell it there must be at least one demand.
- (b) What is your expected revenue from your sale attempt?
- (c) A bike shop offers you 50 TL for this bike. Would you be willing to sell it at that price? (Consider the answer to part b. and explain your answer)
11. You call the call center of a bank and try to gain connection to a call center operator. Assume that all of the operators are busy. You wait on the line until one of the operators becomes available. At each minute, the probability of connecting to a operator is 0.6 and it is independent from the previous minutes. Assume that if lasts non-integer minutes to connect to the operator, your waiting time is taken as the higher nearest integer of this non-integer value. For example, if you gain connection at the 2.5^{th} minute, your waiting time is taken as 3 minutes. Let X be a random variable representing your waiting time.

- (a) What is the probability that you wait on the line less than or equal to 2 minutes?
- (b) What is your expected waiting time?

Assume that your waiting cost is can be expressed in the following way as a function of the waiting time x :

$$C(x) = x^2/3 + 1$$

Note that the unit of the waiting cost is TL.

- (c) What is the probability that your waiting cost is more than 4 TL?
- (d) What is your expected waiting cost?

12. Random variables X and Y have the following joint pmf:

$$p_{X,Y}(x, y) = \begin{cases} \frac{c}{xy}, & \text{if } x \in \{1, 2, 3\} \text{ and } y \in \{1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the coefficient c .
- (b) Find $p_X(x)$ and $E[X]$.
- (c) Let $Z = \max\{X, Y\}$. Find the pmf of Z .
- (d) Find the expected value of Z .
- (e) Find the variance of Z .

13. Consider a random variable X such that

$$p_X(x) = \begin{cases} a, & \text{if } x = 2 \\ b, & \text{if } x = 0 \\ 0.05, & \text{if } x = -1 \\ 0.15, & \text{if } x = 3 \\ 0.25, & \text{if } x = -2 \\ 0, & \text{otherwise.} \end{cases}$$

Given that $E[X] = 0.5$ and $Y = 2X + 1$

(a) Find the value of a and b .

(b) Find $E[X^2 + 2]$.

(c) Find $\text{Var}[Y]$.

(d) Find $P(X \geq Y)$

14. An academic department offers 8 lower level courses: $[L_1, L_2, \dots, L_8]$ and 10 higher level courses: $[H_1, H_2, \dots, H_{10}]$. A valid curriculum consists of 4 lower level courses. and 3 higher level courses.

(a) How many different curricula are possible?

(b) Suppose that $[H_1, \dots, H_5]$ have L_1 as a prerequisite, and $[H_6, \dots, H_{10}]$ have L_2 and L_3 as prerequisites. i.e. any curricula which involve, say, one of $[H_1, \dots, H_5]$ must also include L_1 . How many different curricula are there?